Income Distribution - Problem Set 3

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Question 1 - Variables

You will need to use or construct a variable for hourly wages, and use variables for demographics. In particular, you will need to use information on Total Income, After-tax income, Wages and Salaries, Hours worked, Education, Age, and Gender, as well as the survey weights. List the codes of these variables for your two data sets. Use variables for individuals, not families. Note: Since we will be focussing on wages in this assignment, you do not need to consider business income (differently from the first assignment).

Answer:

Table 1: Variable Name Mapping for SCF and CIS Datasets

Variable Description	**SCF Label (1997)**	**CIS Label**
Total Income	TOTINC	TTINC
After-tax Income	INCAFTTX(INCFTX)	ATINC
Wages and Salaries	WAGSAL	WGSAL
Hours Worked	HRSWRK(USHOURS)	ALHRWK
Education	EDUC(SUMED)	HLEV2G
Age	AGE	AGEGP
Gender	SEX	SEX
Survey Weights	WEIGHT	FWEIGHT

Question 2 - Building hourly wages

(a) Download the consumer price index for the relevant years from Statistics Canada. Plot its evolution. (b) Compute hourly wages for each individual. Deflate the hourly

wage using the CPI (use Statscan's annual CPI). To make wage statistics more informative for us, use 2020 as the base year. Hint: maybe the most straightforward way of doing this is to do it year by year in your do file, with one line of code per year.

Answer:

(a)

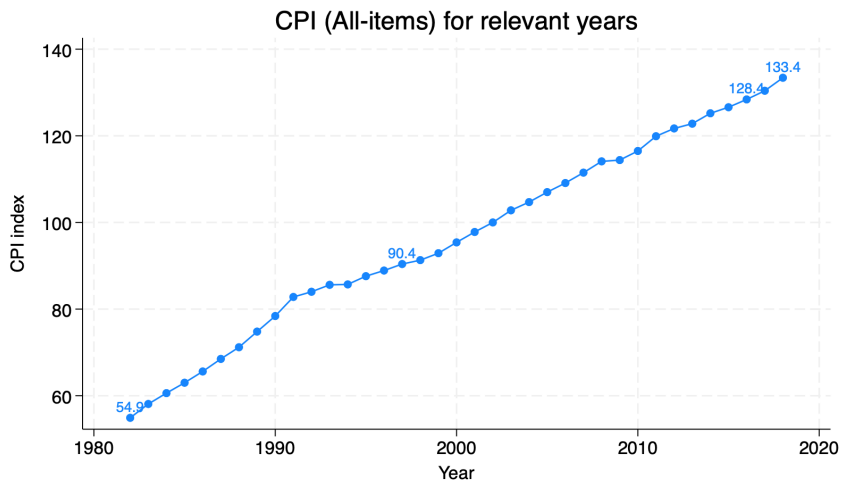

Figure 1: CPI Evolution

Question 3 - Sample selection

Drop from your sample individuals with missing information, zero hours, or a wage below 50% of the minimum wage. Next, keep only individuals between the ages of 25 and 60 (inclusive), who work more than 260 hours. Report in a table like Table 1 of HPV how many observations you have (not using weights) at each stage. Use this selected sample for the remainder of the assignment. Hint: use the yearly minimum wages I provide on mycourses as an Excel file. In principle, you would need to be more precise here, since minimum wages are set at the provincial level in Canada. But to keep things simple, you can use the provided numbers for all individuals. (The file contains the unweighted average of the prevailing minimum wage in Alberta, BC, Ontario and Québec, in current dollars.) Hint 2: In some surveys, age is not

available, only a variable for year of birth, in 5-year intervals. I ask you to use a sample of individuals aged 25 to 60. I suggest that if most individuals in an age group fall in the sample, you keep that age group. For example, those born in 1968 are 25 years old in 1993. You could then include the group of all of those born in 1965-1969.

Answer:

Table 2: Sample selection in the SFC and CIS

	SFC 1982		SFC 1997		CIS 2016		CIS 2018	
	Dropped	Remaining	Dropped	Remaining	Dropped	Remaining	Dropped	Remaining
Initial Sample		79,517		69,461		62,414		94,336
Missing info	41,688	37,829	36,048	33,413	31,287	31,127	47,674	46,662
Wage < 0.5 × min. wage	3,598	34,231	2,642	30,771	1,387	29,740	1,785	44,877
Sample A		**34,231**		**30,771**		**29,740**		**44,877**
Age 25–60	8,285	25,946	4,365	26,406	7,725	22,015	12,562	32,315
Sample B		**25,946**		**26,406**		**22,015**		**32,315**
Annual hours > 260	169	25,777	117	26,289	487	21,528	750	31,565
Sample C		**25,777**		**26,289**		**21,528**		**31,565**

Question 4,5 and 6 - Table

4. *Make sure your data set contains a measure of years of schooling. Round to full years if needed. If there is only categorical information on schooling, convert it to years, assuming that not finishing high school takes 10 years, finishing it takes 12 years, also getting a non-university postsecondary diploma takes 13 years, and getting a university degree takes 16 years.*

5. *Generate a variable potential experience as age minus max (10, years of schooling) minus 6.*

6. *Generate a table with, for each of your samples and years, (a) the mean wage and log variance of wages, (b) mean and variance of years of schooling, (c) mean and variance of experience. How have wage inequality and the inequality of its determinants evolved over time? Does it seem, at this point, as if the evolution of inequality in schooling and experience can explain the evolution of inequality in wages? Here and in the following, do not forget to use weights.*

Answer:

Table 3: Mean wages, log wage variance, schooling, and experience by year

Year	Mean wage	Log var. wages	Mean schooling	Var. schooling	Mean pot. exp.	Var. exp.
1982	24.149	0.309	11.528	8.327	21.164	112.579
1997	27.368	0.336	12.888	4.858	21.035	89.807
2016	33.466	0.354	13.762	6.582	21.395	112.601
2018	37.042	0.361	13.555	8.040	20.684	106.945

Wage inequality, measured by the log variance of wages, rises over time from 0.309 in 1982 to 0.361 in 2018. Average wages also increase steadily across the sample period. This suggests that wages are becoming both higher on average and more dispersed.

Mean years of schooling also rise over time, from 11.528 in 1982 to around 13.5–13.8 in the later years. However, the variance of schooling does not follow the same clear upward trend as wage inequality. It falls between 1982 and 1997, before rising again in 2016 and 2018. While educational attainment increases on average, inequality in schooling does not seem to track wage inequality particularly closely. Possibly due to the education reporting and the difficulty in quantifying it.

Potential experience is fairly constant across years. Its mean remains close to 21 in every sample, while its variance also stays within a similar range. This is not surprising, since potential experience is defined as

$$\text{AGE} - \max(10, \text{Years of schooling}) - 6,$$

so it is mechanically tied to age and schooling. As a result, it would not be expected to change very much over time.

Overall, it does not seem that changes in the inequality of schooling and experience alone can explain the increase in wage inequality.

Question 7 - Mincer the data

7. *For each of your surveys and years, run a basic Mincer wage regression: regress the log wage on years of schooling, as well as potential experience and its square. (a) Report your results in a table with one column for each regression result. Include the number of observations and regression R^2 in the table. (b) What is the marginal return to a year of experience, evaluated at the mean of experience in your latest sample year? (c) Describe your findings. How did the return to schooling or experience change over time? Can these changes explain changes in the evolution of inequality?.*

Answer:

(a)

Table 4: Mincer Wage Regressions by Year

	1982	1997	2016	2018
Years of schooling	0.0583*	0.0838*	0.0684*	0.0539*
	(0.0013)	(0.0016)	(0.0016)	(0.0012)
Potential experience	0.0214*	0.0362*	0.0307*	0.0213*
	(0.0015)	(0.0016)	(0.0016)	(0.0014)
Experience squared	-0.000313*	-0.000516*	-0.000480*	-0.000413***
	(0.000031)	(0.000035)	(0.000037)	(0.000032)
Constant	2.0854*	1.5749*	1.9997*	2.4682*
	(0.0238)	(0.0276)	(0.0272)	(0.0216)
Observations	25,777	26,289	21,528	31,565
R^2	0.0769	0.1154	0.1032	0.0673
Adjusted R^2	0.0768	0.1153	0.1030	0.0672

Notes: Dependent variable is $\ln(WpH)$ (Wage per Hour in 2020 $CAD). Standard errors are reported in parentheses.
* $p < 0.01$.

(b)

Mean of Potential Experience is 21.579819 $\implies$

$$.0213126 + 2(-.0004132)(21.579819) = 00347904$$

(c)

The regressions suggest that the return to schooling rose sharply between 1982 and 1997, then fell back over time. The coefficient on years of schooling increases from 0.058 in 1982 to 0.084 in 1997, before declining to 0.068 in 2016 and 0.054 in 2018. Interpreting this in the usual way, this means the wage premium from an additional year of education was strongest in 1997 and weakest in 2018. So, the evidence points to a rise in the education wage premium during the 1980s and 1990s, followed by a reversal afterwards.

The return to experience shows a similar pattern. The coefficient on potential experience rises from 0.021 in 1982 to 0.036 in 1997, then falls to 0.031 in 2016 and 0.021 in 2018. At the same time, the coefficient on experience squared is always negative, so wages rise with experience, but at a decreasing rate. The implied turning point is about 34.2 years of experience in 1982, 35.1 in 1997, 32.0 in 2016, and 25.8 in 2018, so by 2018 the experience premium flattened out earlier in the lifecycle than in the earlier years.

*** Turning point is calculated as when the marginal effect becomes 0, the classic optimisation.

$$\beta_1 + 2\beta_2 x^* = 0, \quad \Longrightarrow$$

$$x^* = -\frac{\beta_1}{2\beta_2}$$

In terms of inequality, these changes can explain part of the evolution, but not all of it. The rise in the schooling premium from 1982 to 1997 is consistent with higher wage inequality, because workers with more education would pull further ahead of less educated workers. The stronger early-career experience premium in 1997 also helps explain this. However, the decline in both the schooling and experience coefficients by 2016 and especially 2018 suggests that these observed returns became a weaker source of wage differentials later on. So the regressions support the idea that changes in returns to education and experience helped drive inequality upward in the earlier period, but they are unlikely to fully explain the whole story, especially in the later part.

A final point is that the R^2 values remain fairly low, between about 0.07 and 0.12, which means that schooling and experience explain only a modest share of wage variation. That suggests other factors, such as occupation, industry, unions,

technology, institutions, or unobserved skill differences, must also have played an important role in the evolution of inequality.

Question 8 and 9 - Custom Mincers

8. *For each of your surveys and years, run two other Mincer regression specifications of your choice. (Try out additional explanatory variables, different functional forms...) Report results in two tables (one for each specifications, each containing results for all your surveys/years). Briefly comment on why you chose your specifications, which specification you prefer, and why.*

Answer:

Table 5: Mincer Wage Regressions with Gender Interactions

	1982	1997	2016	2018
Years of schooling	0.0599***	0.0746***	0.0596***	0.0491***
	(0.0015)	(0.0021)	(0.0019)	(0.0015)
Potential experience	0.0337***	0.0442***	0.0312***	0.0228***
	(0.0018)	(0.0022)	(0.0023)	(0.0019)
Experience squared	-0.000536***	-0.000678***	-0.000484***	-0.000453***
	(0.000038)	(0.000047)	(0.000051)	(0.000043)
Female	-0.1393***	-0.3356***	-0.5764***	-0.4368***
	(0.0471)	(0.0557)	(0.0538)	(0.0423)
Female × Schooling	0.0075***	0.0212***	0.0267***	0.0158***
	(0.0026)	(0.0033)	(0.0031)	(0.0023)
Female × Experience	-0.0283***	-0.0147***	-0.0034	-0.0016
	(0.0029)	(0.0032)	(0.0033)	(0.0027)
Female × Experience^2	0.000509***	0.000248***	0.000064	0.000031
	(0.000060)	(0.000069)	(0.000073)	(0.000061)
Constant	2.0725***	1.6906***	2.2517***	2.6535***
	(0.0279)	(0.0362)	(0.0345)	(0.0281)
Observations	25,777	26,289	21,528	31,565
R^2	0.1839	0.1536	0.1427	0.1114
Adjusted R^2	0.1837	0.1534	0.1424	0.1112

Notes: Dependent variable is $\ln(WpH)$. Standard errors are in parentheses.
Female is the indicator for women, so the baseline group is men.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Mincer Wage Regressions with Province Fixed Effects

	1982	1997	2016	2018
Years of schooling	0.0568***	0.0799***	0.0643***	0.0522***
	(0.0013)	(0.0016)	(0.0015)	(0.0012)
Potential experience	0.0216***	0.0368***	0.0299***	0.0224***
	(0.0015)	(0.0016)	(0.0017)	(0.0014)
Experience squared	-0.000318***	-0.000552***	-0.000467***	-0.000446***
	(0.000031)	(0.000035)	(0.000037)	(0.000031)
Prince Edward Island	-0.1350***	-0.0144	-0.1118***	-0.1232***
	(0.0273)	(0.0288)	(0.0311)	(0.0246)
Nova Scotia	-0.0757***	-0.0361	-0.0644**	-0.0885***
	(0.0198)	(0.0241)	(0.0257)	(0.0212)
New Brunswick	0.0032	0.0445*	-0.0675***	-0.0947***
	(0.0199)	(0.0243)	(0.0257)	(0.0210)
Quebec	0.1386***	0.1340***	-0.0099	-0.0214
	(0.0174)	(0.0213)	(0.0215)	(0.0178)
Ontario	0.0914***	0.2062***	0.0603***	0.0684***
	(0.0167)	(0.0208)	(0.0210)	(0.0173)
Manitoba	-0.0061	0.0847***	-0.0178	0.0116
	(0.0191)	(0.0234)	(0.0229)	(0.0196)
Saskatchewan	0.0328*	0.0473*	0.1247***	0.1006***
	(0.0191)	(0.0243)	(0.0235)	(0.0199)
Alberta	0.1245***	0.0989***	0.2112***	0.2172***
	(0.0176)	(0.0229)	(0.0225)	(0.0186)
British Columbia	0.1711***	0.2002***	0.0693***	0.0693***
	(0.0181)	(0.0228)	(0.0228)	(0.0188)
Constant	2.0298***	1.4681***	2.0456***	2.4601***
	(0.0274)	(0.0341)	(0.0328)	(0.0266)
Observations	25,777	26,189	21,528	31,565
R^2	0.0957	0.1244	0.1149	0.0902
Adjusted R^2	0.0953	0.1240	0.1144	0.0899

Notes: Dependent variable is $\ln(WpH)$. Standard errors are in parentheses.
Province coefficients are relative to the omitted reference province.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

I estimated two alternative Mincer specifications. First, I added a female dummy and interactions between female and schooling and experience, to allow the returns to human capital to differ by gender. Second, I added province fixed effects, to control for regional wage differences across Canada.

The gender-interaction specification suggests important heterogeneity in wage determination. The female dummy is negative in every year, indicating lower wages for women, while the female-schooling interaction is positive and significant throughout, suggesting a slightly higher return to schooling for women than for men. The female-experience interactions are negative in the earlier years and become small and insignificant by 2016 and 2018, implying that gender differences in experience profiles narrowed over time, which would be consistent with an increased number of women in the workforce. This specification also delivers the largest increase in R^2. At the beginning of the period it is significantly more explanatory than at the end.

The province fixed-effects specification leaves the main schooling and experience coefficients broadly unchanged, showing that the core Mincer results are robust to controlling for geography. Province effects are often sizeable, which confirms that regional wage differences matter. In 1997 it has the largest explanatory value

Overall, I prefer the gender-interaction specification because it is more informative and has greater explanatory power. It captures an important source of wage inequality directly and how it has evolved over time, whereas the province fixed-effects model is perhaps better viewed as a robustness check or an indicator of the poorest provinces.

9. *Write down the regression equation for the basic Mincer wage regression you ran above. Explain, using this equation, why wage changes can be due to changes in attributes of a population, in the prices of/returns to these attributes, and to changes in residuals.*

Answer:

$$\ln w_i = \beta_0 + \beta_1 Schooling_i + \beta_2 Experience_i + \beta_3 Experince_i^2 + \varepsilon_i$$

Wages can change because the attributes of the population change. For example, if later cohorts have more years of schooling, or a different experience profile, then wages may rise even if the coefficients stay the same. In the equation, this is a change in the values of $Schooling_i$ or $Experience_i$. This is often called a composition effect.

Secondly, wages can change because the returns to attributes change. Even if the average level of schooling or experience in the population stays the same, wages will change if the labour market rewards those characteristics differently. In the equation,

this is a change in the coefficients β_1, β_2 and β_3. For instance, if β_1 rises, an extra year of schooling raises wages more than before. This is a price effect or return effect.

Finally, wages can change because the residual changes. The term ε_i includes factors not observed in the regression, such as ability, occupation, firm-specific factors, industry effects, motivation, or measurement error. If the distribution of these unobserved factors changes over time, then wages can change even if both observed attributes and their returns remain unchanged. In particular, changes in the dispersion of residuals can contribute to changes in wage inequality.

Question 10, 11 and 12 - Changes in prices vs changes in attributes.

10. *Return to the basic Mincer wage regression, for all your survey years. What is the explanatory power of the regression? How large is "explained inequality" (inequality in predicted hourly wages) in your four surveys? Does explained or unexplained variance of log wages increase more?.*

11. *What regression should you run so you can easily test if the change in the return to schooling between two surveys is significant? Do it, for your earliest and latest survey.*

12. *Over your survey period, both prices and attributes changed. Imagine, however, that only prices (coefficients) changed, and that attributes were, in all years, as in 2017. How can you generate an estimate of what explained inequality in earlier years would have been under this scenario? Explain, and execute. (Do this using only data for 2017 and your earliest other survey).*

Answer: 10

Explanatory power (R^2):

- 1982 - 0.0769
- 1997 - 0.1154
- 2016 - 0.1032
- 2018 - 0.0673

$$\text{Explained variance} = R^2 \times \text{Total MS}$$
$$\text{Unexplained variance} = (1 - R^2) \times \text{Total MS}$$

Total MS:

- 1982 - .308694645
- 1997 - .336205174
- 2016 - .354372405
- 2018 - .361477537

Table 7: Explained and Unexplained Variance of Log Wages

Year	Explained variance	Unexplained variance
1982	0.0237	0.2850
1997	0.0388	0.2974
2016	0.0366	0.3178
2018	0.0243	0.3372

As we can clearly see unexplained variance increases more.

11

To test whether the return to schooling changed significantly between the earliest and latest survey, I pooled the 1982 and 2018 data, created a year dummy equal to 1 for 2018 and 0 for 1982, and estimated a fully interacted Mincer regression:

$$\begin{aligned}\ln W_i = \beta_0 &+ \beta_1 \text{Schooling}_i + \beta_2 \text{Experience}_i + \beta_3 \text{Experience}_i^2 \\ &+ \delta \text{Year}_i + \gamma_1(\text{Year}_i \times \text{Schooling}_i) \\ &+ \gamma_2(\text{Year}_i \times \text{Experience}_i) + \gamma_3(\text{Year}_i \times \text{Experience}_i^2) + u_i\end{aligned}$$

In this specification, β_1 is the return to schooling in 1982, while γ_1 measures the change in the return to schooling in 2018 relative to 1982. Therefore, testing whether the return to schooling changed over time is equivalent to testing whether $\gamma_1 = 0$.

The estimated coefficient on Year × Schooling is

$$\hat{\gamma}_1 = -0.0061394 \quad (SE = 0.0017577,\ p = 0.000),$$

which is negative and statistically significant at the 1% level. This implies that the return to schooling in 2018 was significantly lower than in 1982.

From the regression, the estimated return to schooling in 1982 is

$$\hat{\beta}_1 = 0.058324,$$

while the estimated return to schooling in 2018 is

$$\hat{\beta}_1 + \hat{\gamma}_1 = 0.058324 - 0.0061394 = 0.0521846.$$

Thus, the return to an additional year of schooling fell from about 5.83% in 1982 to about 5.22% in 2018.

The evidence shows that the return to schooling declined modestly but significantly over the period.

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I am assuming that you mean 2018 instead of 2017.

- $Var(X_{2018}\beta_{1982}) = 0.02819693$
- $Var(X_{2018}\beta_{2018}) = 0.02350201$

This means that if workers had the 2018 distribution of schooling and experience, but wages were priced according to 1982 returns, explained inequality would have been about 0.0282. Under actual 2018 returns, explained inequality is lower, at about 0.0235.

So, holding attributes fixed at 2018 levels, the change in coefficients between 1982 and 2018 reduced explained inequality by about 0.0047. In other words, changes in the returns to schooling and experience made the wage structure associated with observed attributes more equal.

Question 13, 14, 15 and 16 - Residual wage inequality.

13. *Return to the basic Mincer wage regression. Run it again for each survey and year, and save the residuals.*

14. *Report the variance of the residuals. Which fraction of log wage variance does it constitute in each survey?.*

15. *Overall, how do your findings compare to those of AKK?.*

16. *Include some additional variables in your regression. Which ones have strong explanatory power?.*

Answer:13, 14

Table 8: Variance of Residuals and Share of Log Wage Variance

Year	Residual variance	Share of log wage variance
1982	0.28495	0.92309
1997	0.30253	0.89983
2016	0.30602	0.86354
2018	0.32203	0.89088

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My findings are broadly consistent with AKK in showing that observed schooling and experience explain only a limited share of wage inequality, while residual dispersion remains large. This supports the idea that changes in wages are driven by more than just observed human-capital characteristics. However, my results are less detailed than AKK's, because we did not look at upper-tail and lower-tail inequality or formally decompose inequality into price and composition effects. AKK show that price changes remain crucial, especially for upper-tail inequality, whereas the simple Mincer framework only shows that most inequality remains unexplained by schooling and experience alone.

16

The additional variables with the strongest explanatory power are the gender dummy and the gender interactions with schooling and experience. These substantially increase the R^2 , indicating that wages differ not only by education and experience, but also by gender and by the way returns to education and experience vary across gender.

Province dummies also add explanatory power, showing that regional wage differences matter, but their contribution is smaller than that of the gender-interaction specification.